

1 Perfect Secrecy

Definition 1.1: Correctness

$$\forall k \in \text{Im}(\text{Gen}), \forall m \in \mathcal{M}, \\ \text{Dec}(k, \text{Enc}(k, m)) = m$$

Definition 1.2: Shannon Secrecy

An encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is said to be Shannon Secret if for every probability distribution over $\mathcal{M}$, $\forall m \in \mathcal{M}$, and $\forall c \in \mathcal{C}$ for which $\mathbb{P}[c \in \mathcal{C}] > 0$,

$$\mathbb{P}[M = m \mid C = c] = \mathbb{P}[M = m]$$

Definition 1.3: Perfect Secrecy

An encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is said to be Perfectly Secret if $\forall m_0, m_1 \in \mathcal{M}, \forall c \in \mathcal{C}$

$$\mathbb{P}[C = c \mid M = m_0] = \mathbb{P}[C = c \mid M = m_1]$$

Theorem 1.4

Perfect Secrecy is equivalent to Shannon Secrecy

Theorem 1.5

If an encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ is perfectly secret with message space $\mathcal{M}$ and key space $\mathcal{K}$, then $|\mathcal{K}| \geq |\mathcal{M}|$

Definition 1.6: One-Time Pad

$\mathcal{M} = \mathcal{C} = \mathcal{K} = \{0, 1\}^\ell$, where $\ell \geq 1$
 $\text{Gen} : k \xleftarrow{\$} \mathcal{K}$,
 $\text{Enc}(k, m) : c = m \oplus k$,
 $\text{Dec}(k, m) : c \oplus k$

1.1 Flaws with perfect secrecy

- Can't use same key twice.
- Key must be chosen uniformly at random and be at least as long as the message.

2 Negligible Functions

Definition 2.1: Negligible Function

A non-negative function ϵ is negligible if for all $c \geq 0$ there exists a $N \in \mathbb{N}$ s.t. for all $n \geq N$

$$\epsilon(n) \leq 1/n^c$$

Theorem 2.2

For f, g polynomially bounded functions and negligible functions ϵ and ϕ .

- $f \cdot \epsilon = \eta$
- $f + g = h, f \cdot g = h'$
- $\epsilon + \phi = \eta', \epsilon \cdot \phi = \eta''$

For η, η', η'' negligible functions and h, h' polynomially bounded functions.

3 Computationally Secure

Definition 3.1: Computationally Indistinguishable

Given ensembles $\{X_n^0\}_{n \in \mathbb{N}}$ and $\{X_n^1\}_{n \in \mathbb{N}}$ where X_n are distributions over $\{0, 1\}^{\ell(n)}$ where ℓ is some polynomial, we say that $\{X_n^0\}_{n \in \mathbb{N}}$ and $\{X_n^1\}_{n \in \mathbb{N}}$ are computationally Indistinguishable iff $\forall$ PPT $\mathcal{A}$, there exists a $\text{negl}(\cdot)$ s.t. for all $n \in \mathbb{N}$,

(Predictor):

$$\mathbb{P}[b \leftarrow \{0, 1\}, x \leftarrow X_n^b \mid \mathcal{A}(x) = b] \leq 1/2 + \text{negl}(n)$$

(Decider):

$$|\mathbb{P}[x \leftarrow X_n^0 \mid \mathcal{A}(x) = 1] - \mathbb{P}[x \leftarrow X_n^1 \mid \mathcal{A}(x) = 1]| \\ \leq \text{negl}(n)$$

Theorem 3.2

Computationally Indistinguishable ensembles are closed under efficient operations and transitive up to a polynomial amount.

Definition 3.3: EAV Security

$\text{EXPT}_{\text{EAV}}^{b, \mathcal{A}}(1^n)$: $\mathcal{A}$ sends two messages of the same length (m_0, m_1) then the challenger chooses one of the two messages depending on the parameter b then returns c_b to the $\mathcal{A}$. Then $\mathcal{A}$ outputs its guess b' .

4 Pseudorandom Generators

Definition 4.1: PRG

A determ. algo $G : \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)} \in \mathcal{P}$ is a PRG if

- expansion: $\forall n, \ell(n) > n$.
- pseudoran: $\{s \leftarrow U_n \mid G(s)\}_n \approx U_{\ell(n)}$

Definition 4.2: PRG One-Time Pad

Same as One-Time pad but instead of $\oplus k$ do $\oplus G(k)$. Can't reuse key.

5 Multi-msg Security

Definition 5.1: Multi-EAV Security

$\text{EXPT}_{\text{Meav}}^{b, \mathcal{A}}(1^n)$: $\mathcal{A}$ sends two vectors of messages where $|m_{0,i}| = |m_{1,i}|$ for all i . The challenger chooses one of the two vectors and returns one back encrypted. $\mathcal{A}$ outputs its guess b' .

Definition 5.2: Multi-CPA Security

$\text{EXPT}_{\text{LRcpa}}^{b, \mathcal{A}}(1^n)$: $\mathcal{A}$ can query the oracle LR_k^b a polynomial times with m_0^i, m_1^i and each time the oracle will always return either the left message encryption or the right message. Then $\mathcal{A}$ outputs its guess.

Definition 5.3: CPA Security

$\text{EXPT}_{\text{cpa}}^{b, \mathcal{A}}(1^n)$: This basically the Multi-CPA exp. First make queries then input actual challenge it sends back then query then guess.

Theorem 5.4

Multi-CPA $\implies$ Multi-EAV

Multi-CPA $\iff$ CPA

Theorem 5.5

In order for a scheme to be CPA secure, it must have randomized encryption.

6 Pseudorandom Functions

Definition 6.1: PRF

A determ. algo $F : \mathcal{K} \times \{0, 1\}^n \rightarrow \{0, 1\}^n \in \mathcal{P}$ is a PRF if

$$\{k \leftarrow \mathcal{K}, x \leftarrow U_n \mid F_k(x)\} \approx$$

$\{x \leftarrow U_n \mid F(x)\} \approx U_n$ where F is a uniformly random function.

Theorem 6.2

PRG $\iff$ PRF

Definition 6.3: CPA secure with PRF

Let F be PRF.

$$\text{Gen}(1^n) : k \xleftarrow{\$} \{0, 1\}^n$$

$$\text{Enc}(k, m) : r \xleftarrow{\$} \{0, 1\}^n, c = \langle r, F_k(r) \oplus m \rangle$$

$$\text{Dec}(k, \langle r, c' \rangle) : m := F_k(r) \oplus c'$$

7 MAC

Definition 7.1: EU MAC

$\text{EXPT}_{\text{EU}}^{\mathcal{A}}(1^n)$ $\mathcal{A}$ queries then guesses (m, t) . If $\text{Vrfy}(k, m, t) = 1$ and m not queried by $\mathcal{A}$ then output 1.

Definition 7.2: SEU MAC

$\text{EXPT}_{\text{SEU}}^{\mathcal{A}}(1^n)$ Same game but succeeds if the guess (m, t) has not been seen before.

Theorem 7.3

If a MAC with canonical verification EU then it is SEU

$\text{Vrfy}(k, m, t) = 1$ if $\text{MAC}(k, m) = t$ this implies MAC is deterministic

Definition 7.4: PRF MAC

Let F be a PRF.

$$\text{MAC}(k, m) = F_k(m) = t$$

$$\text{Vrfy}(k, m, t) = \text{Canonical}$$

Definition 7.5: MAC for Arb len

Let MAC' be secure for n -bit msg.

$\text{MAC}(k, m)$: part $m = m_1 || m_2 || \dots || m_d$, where $|m_i| = n/4$. Let $r \xleftarrow{\$} U_{n/4}$, then let $t_i = \text{MAC}'(k, r || i || m_i)$ then let $t = (r, t_1, \dots, t_d)$.

8 Hash Functions

Definition 8.1: Hash Family

A determ. algo $H : \mathcal{K} \times \{0, 1\}^* \rightarrow \{0, 1\}^n$ s.t. $* = \text{Arb len}$.

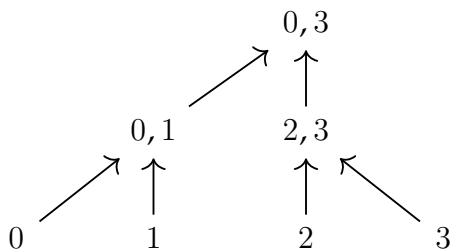
Definition 8.2: Collision Resistant

$\text{EXPT}^{\mathcal{A}}(1^n)$ The challenger sends the key s then the has polynomial time to send back (x, y) s.t. $H^s(x) = H^s(y)$. succeeds if it finds a pair.

Definition 8.3: Hash MAC

Let MAC' be a secure n -bit msg.

$\text{MAC}((k, s), m) = \text{MAC}'(H^s(m))$



Merkle path is the nodes you needed to construct the root.

User only has to store the root digest to verify all of tree. To update a block for example block 1 all the server has to send to you is block 0 and 2,3. Also, same with verifying a block all you need is the block 1, 0, 2,3 and the user stores the root, so they can verify.

To check and make more efficient first make it, so you can lose half the blocks to then lose data then use a probabilistic check to get a probability of $1/2^{-k}$ of passing if they did change one block.

9 Number Theory

Theorem 9.1: Fermat's Little Thm

If p is prime then for $a \in \{1, \dots, p-1\}$ then $a^{p-1} \equiv 1 \pmod{p}$

Example: Pick Subgroups of Prime Order

Let $p = rq + 1$ where p, q are primes.

Then $\mathbb{G} = \{[h^r \pmod{p}] \mid h \in \mathbb{Z}_p^*\}$ is a subgroup of order q .

10 Key Exchange

Definition 10.1: Dlog

$\text{Dlog}^{\mathcal{A}, \mathcal{G}}(1^n)$ Challenger sends $(\mathbb{G}, q, g, h)$ where $h = g^x$ and $\mathcal{A}$ guesses x . If correct then return 1.

Definition 10.2: DDH

$\text{DDH}_b^{\mathcal{A}, \mathcal{G}}(1^n)$ Challenger sends $(\mathcal{G}, q, g, g^x, g^y, h_b)$ s.t. $h_0 = g^z$ and $h_1 = g^{xy}$. Then $\mathcal{A}$ guesses if h_b is random or g^{xy} .

DDH is harder than Dlog.

Definition 10.3: DH Key Exchange

First person samples a prime order multiplicative group $(\mathbb{G}, q, g)$ and samples an $x \in \mathbb{Z}_q$ then they send $(\mathbb{G}, q, g), g^x$ to the second person. The second person then samples $y \in \mathbb{Z}_q$ and then sends g^y back then they both can compute g^{xy} .

Definition 10.4: Key Expt

$\text{KEEXPT}_b(1^n)$ Two parties run the key exchange protocol. Let Γ be the transcript of the protocol and k the outputted key at the end. If $b = 0$ output γ, k and else γ, k' where $k' \xleftarrow{\$} \mathcal{K}$.

11 Public-Key Crypt

11.1 Syntax

$\text{Gen}(1^n) \rightarrow (pk, sk) \text{ Enc}(pk, m) \rightarrow c \text{ Dec}(sk, c) = m$

Definition 11.1: Correctness

Except with negligible probability over the choice of (pk, sk) output by $\text{Gen}(1^n)$, it must be that for any legal msg m ,

$$\text{Gen}(sk, \text{Enc}(pk, m)) = m$$

Definition 11.2: EAV

$\text{PKEXPT}_{\text{eav}}^{b, \mathcal{A}}(1^n)$ Challenger sends a public key pk and then $\mathcal{A}$ sends m_0, m_1 and challenger picks one to encrypt and send back.

Theorem 11.3

A public key encryption protocol is EAV-secure if and only if it is CPA-secure.
This implies it must be random.

Definition 11.4: El Gamel Enc

Run the group protocol to get a prime order multiplicative group $(\mathbb{G}, q, g)$.

Gen: sample $x \leftarrow \mathbb{Z}_q$ and output $pk = (\mathbb{G}, q, h = g^x)$ and $sk = (\mathbb{G}, q, g, x)$.

Enc(pk, m): choose random $y \leftarrow \mathbb{Z}_q$ and output (g^y, mh^y)

Dec($sk, (c_1 = g^y, c_2 = mh^{xy})$): $c_2 \cdot g^{-xy} = m$

Definition 12.2: Signature EU

$\text{EXPT}^{\mathcal{A}}(1^n)$ Challenger sends public key. $\mathcal{A}$ makes queries. Then sends its guess (m, σ) and succeeds if $\text{Vrfy}(pk, m, \sigma) = 1$ and m was not queried before.

Definition 12.3: Factoring

$\text{Factor}^{\mathcal{A}}(1^n)$ Sample a random $N = pq$, where p, q are n -bit primes. $\mathcal{A}$ is given N and outputs two primes p', q' . Succeeds if $p'q' = N$.

Definition 12.4: RSA

$\text{RSA}^{\mathcal{A}}(1^n)$ Sample a random $N = pq$, where p, q are n -bit primes. Sample e, d s.t. $ed = 1 \mod \phi(N) = (p-1)(q-1)$ Sample $y \in \mathbb{Z}_N^*$ then $\mathcal{A}$ sends back an x succeeds if $x^e = y \mod N$.

Definition 12.5: Hash, RSA, and Sign

Gen sample a random $N = pq$ s.t. p, q are n -bit primes. Sample e, d s.t. $ed = 1 \mod \phi(N) = (p-1)(q-1)$. $sk = (N, d)$, $pk = (N, e)$ and generate a hash function.

Sign(sk, m) = $(H(m))^d \mod N = \sigma$

Vrfy(pk, m, σ) and output 1 if $H(m) = \sigma^e \mod N$.

12 Signature

12.1 Syntax

$\text{Gen} \rightarrow (pk, sk) \text{Sign}(sk, m) \rightarrow \sigma, \text{Vrfy}(pk, m, \sigma) \rightarrow b$

Definition 12.1: Correctness

Except with negligible probability over the choice of (pk, sk) output by $\text{Gen}(1^n)$, it must be that for any legal msg m ,

$$\text{Vrfy}(pk, m, \text{Sign}(sk, m)) = 1$$